

SHIVANSH UPADHAYAY

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PROFESSIONAL SUMMARY

Full-Stack and AI/ML Engineer with strong foundations in OOP and DSA (250+ LeetCode, 95.67 percentile Naukri Young Turks), with 3 production-grade applications shipped end-to-end. Built MedLens (YOLOv8 + BioBERT + RAG + Gemini Flash multilingual diagnostic interpreter), Astra AI (6-agent self-correcting LangGraph platform with Groq LLaMA 3.3 70B), and BookMyStay (MERN hotel booking with Razorpay + RBAC). Applied ML to 7,995 sq km of satellite imagery at Indian Space Academy. Targeting a Software Engineering Internship to build scalable, intelligent products.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, NoSQL, HTML5, CSS3

Core CS: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), System Design Fundamentals

Frontend / Backend: React.js, Tailwind CSS, Axios | Node.js, Express.js, REST APIs, JWT, Prisma ORM

Databases: MongoDB, MongoDB Atlas, PostgreSQL, MySQL

AI / ML: LangGraph, LangChain, YOLOv8, BioBERT, Pinecone, RAG, Groq API (LLaMA 3.3 70B), Gemini Flash, Streamlit, Tavily, Google Earth Engine

Cloud & DevOps: AWS, Docker, Git, GitHub, Vercel, Render | Razorpay API Integration

PROJECTS

MedLens – AI Diagnostic Report Interpreter

2026

Stack: YOLOv8, BioBERT, Pinecone, RAG Pipeline, Gemini Flash, Python, Streamlit

- Designed an AI platform interpreting medical lab reports and scans for rural patients — YOLOv8 for imaging anomaly detection, BioBERT for medical NER, Pinecone-backed RAG over clinical literature, and Gemini Flash for multilingual output (Hindi, Tamil, Bengali, and more).
- Enables zero-friction diagnostic access for patients with no medical background, removing language and literacy barriers in underserved regions.

Astra AI – Agentic Multi-Agent Research Platform

April 2026

Stack: LangGraph, LangChain, Groq API (LLaMA 3.3 70B), Streamlit, Tavily, FPDF2, BeautifulSoup4

- Built a 6-agent autonomous research platform with a Critic-driven Reflection-Retry loop — scores outputs (0–10) and auto-retries up to 3× below 8/10 — featuring LangGraph stateful routing, Mermaid.js flowcharts, conversational memory, and MD/PDF export.
- Reduced research synthesis from hours to under 5 minutes via Tavily live search and BeautifulSoup4 web extraction, powered by Groq's LLaMA 3.3 70B.

BookMyStay – Full-Stack Hotel Booking Platform

December 2025

Stack: React.js, Node.js, Express.js, MongoDB, Prisma ORM, Tailwind CSS, JWT, Razorpay API

- Built a hotel booking platform with JWT-enforced RBAC (Admin / Owner / Customer), Razorpay payment gateway, real-time availability search, advanced filters, and automated audit logging.
- Designed a normalized Prisma ORM schema supporting multi-property owners and dynamic pricing; deployed on Vercel (frontend) and Render (backend) with RESTful APIs.

WORK EXPERIENCE

Remote Sensing and GIS Intern | Indian Space Academy (Remote)

January 2026

- Built an AI-enabled forest cover change detection system using Google Earth Engine, NDVI differencing, and adaptive thresholding across 7,995 sq km — quantifying 309.73 sq km forest loss, 1,768.63 sq km stable, and 343.73 sq km gain.

EDUCATION

B.Tech in Computer Science | Rishihood University (Newton School of Technology), New Delhi

Expected 2028

Coursework: OOP, DSA, DBMS, Operating Systems, Software Engineering, Computer Networks

CERTIFICATIONS & ACHIEVEMENTS

- Winter Training Programme on Remote Sensing & GIS – Indian Space Academy, Jan 2026. Satellite image processing, GIS tools, spatial data analysis.
- Merit Certificate – Naukri Campus Young Turks, Oct 2025. 95.67 percentile across 200,000+ participants in Logical Reasoning, Quantitative Aptitude, Verbal Ability, and Problem Solving.
- LeetCode: 250+ DSA problems solved (shivansh_009) — arrays, dynamic programming, graphs, trees, and OOP design patterns.